

EXPLORE → SIMULATE → INVEST → COMPARE

with ♦ Ask DALOY available everywhere.

Water Economics principle	Where your system addresses it
Scarcity	Water Pool, supply shocks, demand projection
Common-pool resource	Shared Resource Pressure
Allocation	Sector Allocation Simulator
Demand management	Tariff, subsidy, NRW, conservation assumptions
Affordability	Household affordability analysis
Valuation	Cost/m ³ , beneficiaries, Public Benefit per Peso
Investment	Project comparison
Financing	Grant/LGU/loan mix
Trade-offs	Baseline vs Scenario
Decision support	Ask DALOY

1. EXPLORE

Purpose

Give the user a fast visual understanding of:

“What is the municipality's current water-economic condition?”

This is your entry screen.

1A. Municipality / Water Economy Map

User sees

A visual map of Samar or selected Region VIII pilot areas.

The user clicks:

CATBALOGAN

The map zooms/selects it and loads its local water system.

Do not open a dashboard full of cards.

Instead show one compact status strip:

CATBALOGAN

💧 Water Stress: **HIGH**

₱ Affordability: **MODERATE**

♻️ NRW: **28%**

🌊 Source Pressure: **92%**

Core input data

municipality

population

households

grossSupply

totalDemand

nrwRate

reserveRate

tariff

householdIncome

sourceCapacity

sectorDemand[]

Data status

Every number should carry one small label:

ACTUAL

HISTORICAL

ESTIMATED

USER INPUT

SIMULATED

PROJECTED

This becomes important because you will mix real Samar data with hackathon assumptions.

2. WATER POOL VISUALIZATION

Purpose

Make **scarcity visible without needing a long explanation**.

This should be the visual centerpiece of the application.

Core calculation

First calculate NRW volume:

$$NRWVolume = GrossSupply \times NRWRate$$

Then reserve:

$$ReserveVolume = GrossSupply \times ReserveRate$$

Then:

$$UsableWater = GrossSupply - NRWVolume - ReserveVolume$$

Instead of one potentially confusing signed value, calculate both:

$$Deficit = \max(0, TotalDemand - UsableWater)$$

$$Surplus = \max(0, UsableWater - TotalDemand)$$

Example

Gross supply:

10,000 m³/day

NRW:

25% = 2,500 m³/day

Reserve:

500 m³/day

Usable:

7,000 m³/day

Demand:

9,000 m³/day

Deficit:

2,000 m³/day

Visual behavior

Use a transparent **2.5D reservoir/pool**.

Water comes from source icons:

 Spring

 Well

 Surface source

↓

enters the municipal pool

↓

flows to:



while NRW appears as a visible **leak**.

Dynamic behavior

Supply ↓

→ water level falls.

NRW ↑

→ leak grows.

Demand ↑

→ sector withdrawal flows widen.

NRW ↓

→ leak shrinks and usable-water level increases.

The numbers support the animation; the animation should be what the judge notices first.

3. SHARED RESOURCE PRESSURE

Water Economics concept

Common-Pool Resource

Purpose

Answer:

“Are all users collectively demanding more than the shared source can support?”

Keep this inside Explore.

MVP calculation

$$ResourcePressure = \frac{RequestedWithdrawal}{AvailableSourceCapacity} \times 100$$

For a municipality using several sources:

$$CombinedCapacity = \sum SourceCapacity_i$$

Use this value unless you possess verified sustainable-yield estimates.

Example

Available source capacity:

8,000 m³/day

Combined demand/withdrawal request:

8,800 m³/day

\[ResourcePressure=110\% \]

States

 **<80% — Normal**

 **80–100% — High Pressure**

 **>100% — Overdrawn / Demand Exceeds Capacity**

Make thresholds editable if desired rather than presenting them as universal regulation.

Visual

Use a ring surrounding the water source:

92% SOURCE PRESSURE

When users change demand later in SIMULATE, this same ring reacts.

4. SIMULATE

Purpose

Answer:

“What happens if the LGU changes a water policy or condition?”

This is your **main interactive workspace**.

The user should be changing numbers and watching the water economy react immediately.

4A. Supply Scenario Control

Inputs

Quick presets:

Normal

–10% Supply

–20%

–30% Drought

Source Failure

plus:

Custom Supply: ____ m³/day

Calculation

For percentage shock:

$$ScenarioSupply = BaselineSupply(1 + SupplyChange)$$

Example:

10,000 m³/day

with –30%:

$$10,000(0.70) = 7,000$$

Visual effect

Immediately update:

- pool level
- deficit
- resource pressure
- sector satisfaction
- allocation
- economic exposure

One input should trigger the **whole model**.

5. NRW SIMULATOR

Purpose

Show the economic value of recovering water that is already being produced.

Inputs

Baseline:

30% NRW

Scenario:

20% NRW

Formula

$$OldNRWVolume = Supply \times OldNRWRate$$

$$NewNRWVolume = Supply \times NewNRWRate$$

$$RecoveredWater = OldNRWVolume - NewNRWVolume$$

Example:

$$10,000 \times (30\% - 20\%)$$

=

1,000 m³/day recovered

Outputs

Show only a few strong outputs:

+1,000 m³/day usable water

–10 percentage points NRW

Estimated water/revenue value: ₱X

Only calculate peso value if a valid tariff/cost assumption exists.

Visual

Make the leak physically shrink.

That visual alone can communicate the feature to judges.

6. SECTOR ALLOCATION SIMULATOR

Water Economics concept

Scarcity + Allocation + Opportunity Cost

Purpose

Answer:

“When there isn't enough water for everybody, how does changing allocation affect different users?”

MVP sectors

Keep it manageable:



Households



Critical/Public Services

 **Agriculture**

 **Fisheries/Aquaculture**

 **Commercial + Tourism**

 **Industry**

You could combine Commercial and Tourism for Level 1 and separate them later.

Each sector stores

name

demandM3

allocatedM3

priorityWeight

minimumProtectedM3

isProtected

Satisfaction

$$Satisfaction_i = \min \left(100, \frac{Allocated_i}{Demand_i} \times 100 \right)$$

Output

 Household **100%**

 Agriculture **67%**

 Fisheries **73%**

 Commercial **78%**

Interaction

Use:

- sliders
- water wedges
- draggable allocation handles

Whenever total supply is constrained:

increasing one allocation must reduce something elsewhere.

That is critical.

9. TARIFF + AFFORDABILITY + DEMAND MANAGEMENT

One feature, not three pages

Purpose

Answer:

“Can the water system remain financially sustainable without making water unaffordable?”

Inputs

currentTariff

proposedTariff

monthlyConsumption

averageIncome

lowIncomeEstimate

demandResponseAssumption

For the simplest MVP, use an **effective ₱/m³ tariff**.

A full progressive-block tariff editor can be a Level 2 improvement.

Monthly bill

Simplified:

$$MonthlyBill = MonthlyConsumption \times Tariff$$

If you implement minimum/fixed charges:

$$Bill = FixedCharge + (VariableConsumption \times Rate)$$

Affordability

$$AffordabilityRatio = \frac{MonthlyWaterBill}{MonthlyHouseholdIncome} \times 100$$

Display

Don't claim one universal threshold.

Allow configurable policy thresholds:

 Low

 Moderate

 High

Display:

3.4% of household income

alongside the risk class.

10. DEMAND-RESPONSE ASSUMPTION

Important:

Do **not** code:

Tariff increases 10% → Demand falls 10%.

Instead:

User enters

Expected Demand Response: -4%

Label:

USER ASSUMPTION

Then:

$$AdjustedDemand = BaselineDemand(1 + DemandResponse)$$

Example:

10,000 m³/day

× 0.96

=

9,600 m³/day.

This allows experimentation without falsely presenting an elasticity estimate as fact.

11. SUBSIDY SIMULATOR

Keep inside Tariff & Affordability

Purpose

Show whether financial assistance can protect vulnerable users while preserving the broader tariff structure.

Inputs

subsidyBudget

targetHouseholds

subsidyPerHousehold

Simple MVP calculation

$$\lfloor \text{CoveredHouseholds} = \min \left(\text{TargetHouseholds}, \left\lfloor \frac{\text{Budget}}{\text{SubsidyPerHousehold}} \right\rfloor \right) \rfloor$$

$$TotalProgramCost = CoveredHouseholds \times SubsidyPerHousehold$$

Then:

$$AdjustedBill = \max(0, OriginalBill - Subsidy)$$

Recalculate affordability.

Visual

Before:

● 5.1%

Apply subsidy

↓

After:

● 3.8%

Then:

1,620 households assisted

Very easy to demonstrate.

12. SIMPLE DEMAND PROJECTION

Purpose

Answer:

“If population and economic demand continue changing, will present supply still be adequate?”

Don't call it advanced forecasting.

Call it:

Demand Projection

Population projection

$$P_t = P_0(1 + g)^t$$

Residential demand

If per-capita demand is expressed in L/person/day:

$$ResidentialDemand = \frac{Population \times PerCapitaDemand}{1000}$$

to produce m³/day.

Then:

$$FutureTotalDemand = ResidentialDemand + Agriculture + Fisheries + Commercial + Tourism$$

Sector values can also have editable growth assumptions.

UI

Timeline:

2026 —————●————— **2035**

Move to 2030.

Population changes.

↓

Demand changes.

↓

Water pool changes.

↓

Shortage risk changes.

Everything remains connected.

13. INVEST - AI suggestion for the meantime

Purpose

Answer:

“Where should the next peso go?”

Do not implement 20 projects.

Use four representative interventions:

 **Pipeline Rehabilitation**

 **NRW Reduction Program**

 **New Water Source**

 **Storage Expansion**

Each uses identical data fields.

```
Project {  
  id  
  name  
  capex  
  annualOpex  
  usefulLifeYears  
  waterGainM3PerDay  
  householdsBenefited  
  reliabilityScore  
  equityScore  
  economicBenefitScore  
  implementationTime  
}
```

14. INTERVENTION ECONOMIC VALUATION - AI suggestion for the meantime

Purpose

Compare projects using their actual economic consequences rather than simply:

“Which creates the most water?”

A. Simple lifecycle cost

For the hackathon:

$$LifecycleCost = CAPEX + (AnnualOPEX \times UsefulLife)$$

You can introduce discounted NPV later.

B. Cost per additional/recovered m³

$$LifetimeWater = WaterGainPerDay \times 365 \times UsefulLife$$

$$CostPerM3 = \frac{LifecycleCost}{LifetimeWater}$$

Label:

Simple lifecycle cost / m³

so judges know it's not a full discounted economic appraisal.

C. CAPEX per beneficiary

$$CAPEXPerHousehold = \frac{CAPEX}{HouseholdsBenefited}$$

D. Economic loss avoided

Where a defensible peso coefficient exists:

₱X/year avoided

Otherwise:

Economic Exposure Reduced: HIGH

Never invent monetary valuation.

15. PUBLIC BENEFIT PER PESO

Purpose

Rank interventions using both economics and public value.

I recommend changing the earlier formula slightly.

Don't subtract raw project cost from unrelated scores.

Instead normalize every dimension to **0–100**.

Subscores

W = Water benefit

E = Equity

L = Livelihood/Economic benefit

R = Reliability

C = Cost efficiency

Then:

$$Score = w_W W + w_E E + w_L L + w_R R + w_C C$$

All weights sum to 1 or 100%.

Example:

Water benefit — **30%**

Equity — **25%**

Economic/livelihood — **20%**

Reliability — **15%**

Cost efficiency — **10%**

Result:

88 / 100

Important

Lower project cost should translate into a **higher Cost Efficiency subscore**.

Do not feed raw pesos directly into the score.

Editable weights

Allow:

⚙️ Policy Weights

Then display:

CUSTOM POLICY WEIGHTS

This makes the system adaptable rather than claiming one universal definition of “best.”

18. TARIFF PRESSURE FROM FINANCING

This is important because it connects **financing back to affordability**.

Calculate approximate annual incremental requirement:

$$IncrementalRequirement = AnnualDebtPayment + AnnualProjectOPEX - AnnualSavings$$

Then:

$$TariffPressure = \frac{IncrementalRequirement}{AnnualBilledVolume}$$

Output:

Estimated additional revenue requirement: +P1.70/m³

Then feed the proposed tariff change into the Affordability Engine.

That creates a complete economic chain:

Project → Financing → Tariff → Household affordability

19. COMPARE

Purpose

Answer:

“Did the scenario actually improve things?”

This should be the final workspace and your presentation payoff.

Use:

BASELINE

versus

SCENARIO

Avoid another giant dashboard.

Show 6–8 major outcomes.

WATER

Deficit

NRW

Source Pressure

PEOPLE

Essential Demand Satisfied

Affordability

ECONOMY

Cost/m³

Benefit/Peso

Tariff Impact

Example:

Indicator	Baseline	Scenario
Water deficit	2,500	900 m ³ /day
NRW	30%	20%
Source pressure	108%	93%
Essential demand met	84%	100%
Low-income affordability	High	Moderate
Cost/m ³ gained	—	₱X
Benefit/Peso	—	88/100
Tariff pressure	—	+₱1.20/m ³

Make this primarily a **visual before/after split**, with the table as supporting detail.

20. DECISION SNAPSHOT

Don't make another page.

At the bottom of COMPARE:

Decision Snapshot

Strongest current trade-off: NRW Reduction + Pipeline Rehabilitation

Why

 More recoverable water per peso

 Essential demand reaches 100%

₱ Lower tariff pressure than new-source alternative

 **Trade-off**

Requires upfront investment and implementation time.

Confidence

SIMULATED / MODERATE

This can be generated from the calculation engine and then rewritten concisely by AI.

21. AI DECISION ASSISTANT

◆ Ask DALOY

This should be **global**.

No separate AI page.

Use a small floating button in every workspace.

Purpose

The AI does not solve the mathematics.

It helps answer:

“What does this result mean, what trade-off am I creating, and what should I test next?”

AI RULE 1 — Calculations happen BEFORE AI

Architecture:

User Inputs

↓

Calculation Engines

↓

Structured Results

↓

Ask DALOY

↓

Explanation / Suggestions

Never:

User Inputs → AI invents calculation

AI RULE 2 — Give it structured context

Example:

```
{
  "municipality": "Catbalogan",
  "scenario": "Drought + NRW Reduction",
  "supply": 8000,
  "demand": 9500,
  "nrw": 20,
  "deficit": 900,
  "sourcePressure": 93,
  "sectorSatisfaction": {
    "households": 100,
    "agriculture": 67,
    "commercial": 76
  },
  "affordability": {
    "lowIncome": "moderate"
  },
  "investment": {
    "project": "Pipeline Rehabilitation",
    "cost": 18000000,
    "waterGain": 850,
    "benefitScore": 88
  }
}
```

}

Also pass data-status labels.

AI RULE 3 — Four jobs only

Explain

Why did the water deficit improve?

Compare

Why does NRW reduction rank above the new source?

Suggest

What scenario should we test next?

Warn

Who becomes disadvantaged by this allocation?

That's enough for the MVP.

22. AI QUICK ACTIONS

Under the AI bubble:

Explain this result

Who is disadvantaged?

Which option gives better value?

What should I test next?

This eliminates the need for users to write complicated prompts.

23. AI RECOMMENDATION FORMAT

Keep every response compact:

◆ DALOY Insight

Suggested option

NRW Reduction + Pipeline Rehab

Why

- lower lifecycle cost/m³
- more essential demand protected
- lower financing pressure

Trade-off

Agricultural demand remains only 78% satisfied.

Next test

Try increasing agriculture allocation by 5% while preserving household minimums.

This perfectly matches your preference for minimal lengthy text.

24. AI SHOULD BE ABLE TO PROPOSE — BUT NOT SECRETLY APPLY — CHANGES

A useful addition:

AI says:

“Try reducing NRW to 20% and using 40% grant financing.”

Then show:

[Try Suggested Scenario]

Clicking it fills the proposed inputs.

But then the **calculation engine reruns everything**.

Only after that may the AI explain the new results.

This keeps AI useful without giving it uncontrolled decision power.

Final visible navigation

You don't need 16 tabs.

Use only:

EXPLORE

Map

Water pool

Shared-resource pressure

SIMULATE

Supply

NRW

Allocation

Tariff

Affordability

Subsidy

Demand projection

INVEST

Projects

Economic valuation

Benefit/Peso

Financing

COMPARE

Baseline vs Scenario

Decision Snapshot

Always visible:

⚙ **Assumptions**

and:

◆ **Ask DALOY**

Final unified logic

The entire system should function as one chain:

Select LGU

↓

See finite water

↓

Identify scarcity

↓

Change supply / demand / tariff / NRW / allocation

↓

Watch people and sectors react

↓

Choose an intervention

↓

Measure its economic value

↓

Build its financing mix

↓

See tariff and affordability consequences

↓

Compare with baseline

↓

Ask DALOY to explain the trade-offs